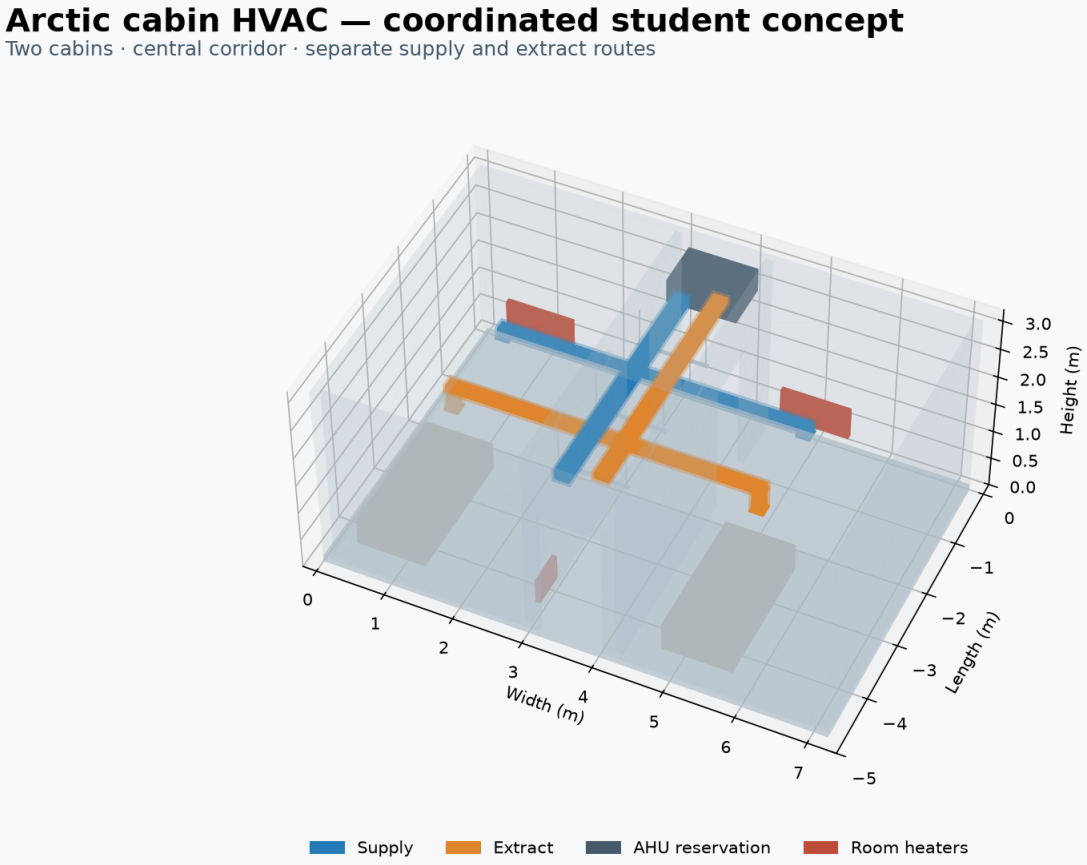
# Arctic cabin HVAC concept

Design study: two cabins, central corridor and separate supply / extract routes




## Purpose

Explore heating demand and spatial coordination for a small accommodation module inspired by Arctic shipbuilding.

## Deliverables

Dimensioned plan and section; preliminary heat balance; routing comparison; source model and calculation files.

## Boundaries

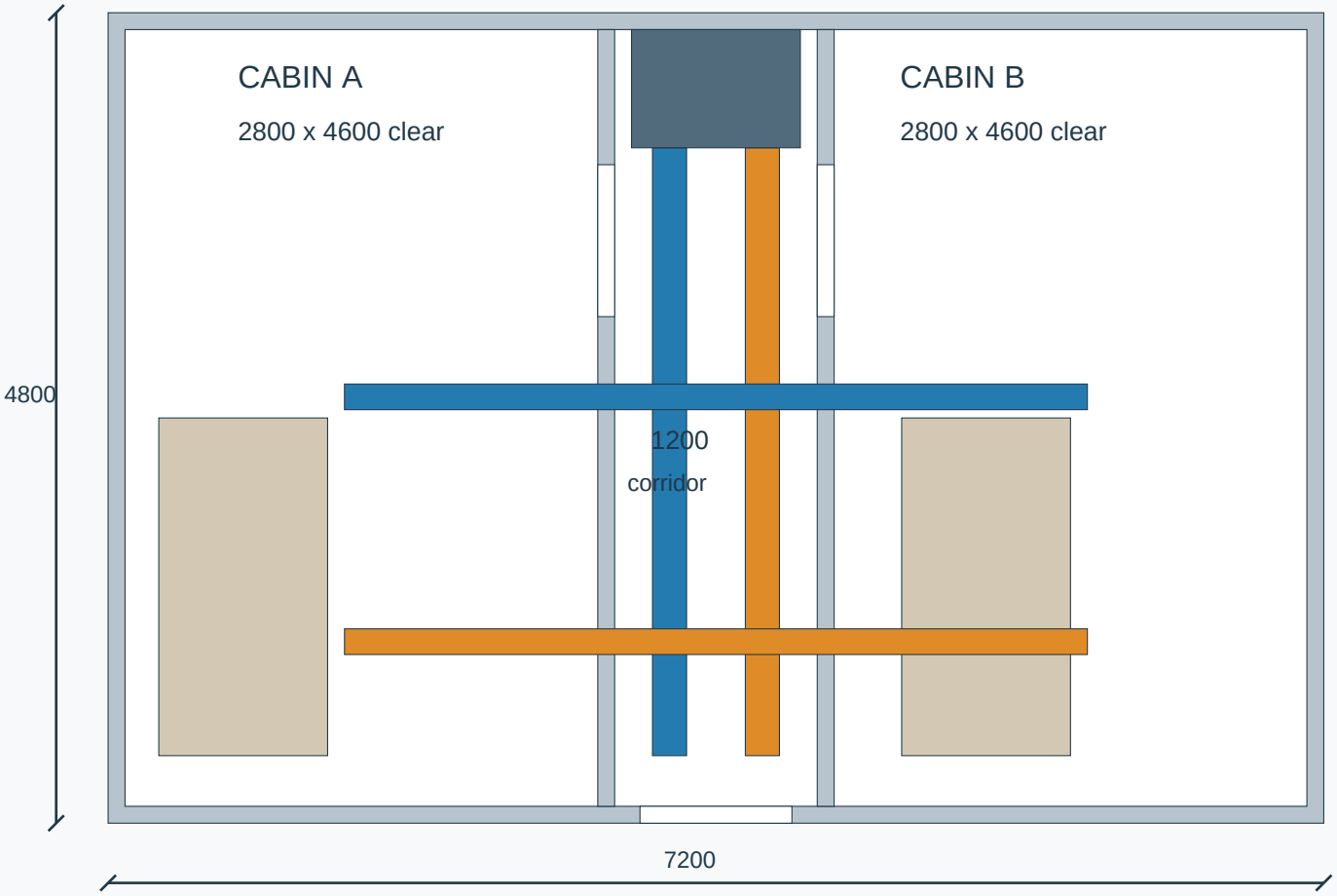
Synthetic geometry and assumed thermal inputs. This is not a drawing of an actual polar icebreaker.

## Authorship

AI-assisted calculation and CAD drafting. Review the assumptions and reproduce the key calculations before presenting this as your project.

# D-01 | General arrangement and duct plan

Dimensions in mm | Plan diagram derived from model coordinates | NTS - use stated dimensions




Partitions: X=2900-3000 and 4200-4300; 100 thick.

Cabin door openings: 900 wide. Entry: 900 wide. Door swings not resolved.

## SUPPLY - BLUE

40 L/s total; 20 L/s per cabin. Main 200 x 150 clear; branches 150 x 100 clear.

## EXTRACT - ORANGE

40 L/s total; 20 L/s per cabin. Main 200 x 100 clear; branches 150 x 100 clear.

## PLAN PROJECTION

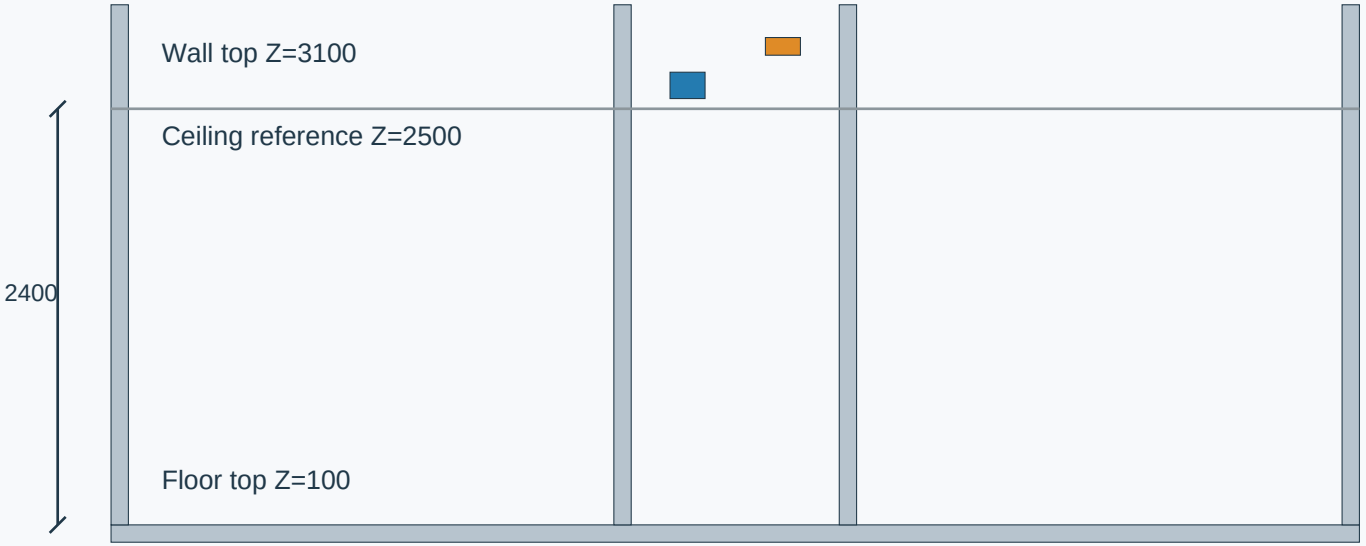
Branch crossings are at different heights. Refer to section. Supply and extract terminate at a generic AHU reservation.

## LAYOUT LIMIT

Module interface ends at the AHU. Intake/exhaust weather protection belongs to ship-level design. Bunks, heaters and AHU are generic reservations.

# D-02 | Vertical coordination section

Across module | Model elevations from bottom of floor reference | NTS




## DUCT ELEVATIONS

Supply main: Z=2559 to 2711. Extract main: Z=2809 to 2911. Vertical service allowance: 600 above occupied ceiling.

25 mm insulation envelopes are modeled. Minimum insulated duct clearance to the roof reference is 164 mm; supply/extract envelope gap at the crossing is 48 mm.

Revision B raises wall height to 3000 mm above the floor. The occupied ceiling stays at 2400 mm. Three rail/rod support sets are concept reservations, not structurally rated.

## Coordination hold points

Support rails at model Y=-4100, -3000 and -1700. Rails reserve 1000 x 40 x 30 mm; rods reserve 12 x 12 mm. Structural attachment, loads and vibration sizing are outside this concept study.

Reserve a removable 1000 x 700 ceiling panel below the AHU; maintain the corridor below as the access route. Actual equipment governs final clearances.

# C-01 | Preliminary heating and airflow

Steady-state sensible heating | Design case: 22 C indoors / -30 C outdoors

Q transmission = (0.30 x 34.56 + 0.35 x 21.60) x 52 = 932 W

Q infiltration = 1.2 x 1005 x (75.072 x 0.15 / 3600) x 52 = 196 W

Q fresh air = 1.2 x 1005 x 0.040 x 52 = 2508 W

Total = 3637 W; with separate 15% sizing allowance = 4182 W

## Zone allocation (preliminary gross-area method)

| Zone | Roof m2 | End wall m2 | Local heat W | With 15% W |
|---|---|---|---|---|
| Cabin A | 14.16 | 8.85 | 463 | 532 |
| Corridor | 6.24 | 3.90 | 203 | 233 |
| Cabin B | 14.16 | 8.85 | 463 | 532 |

The fresh-air coil separately needs 2.88 kW including allowance. Air is supplied at room temperature, so it does not cover local losses. Capacity targets: 0.60 kW per cabin and 0.30 kW for the corridor. The corridor heater is now reserved in CAD; actual products remain unselected.

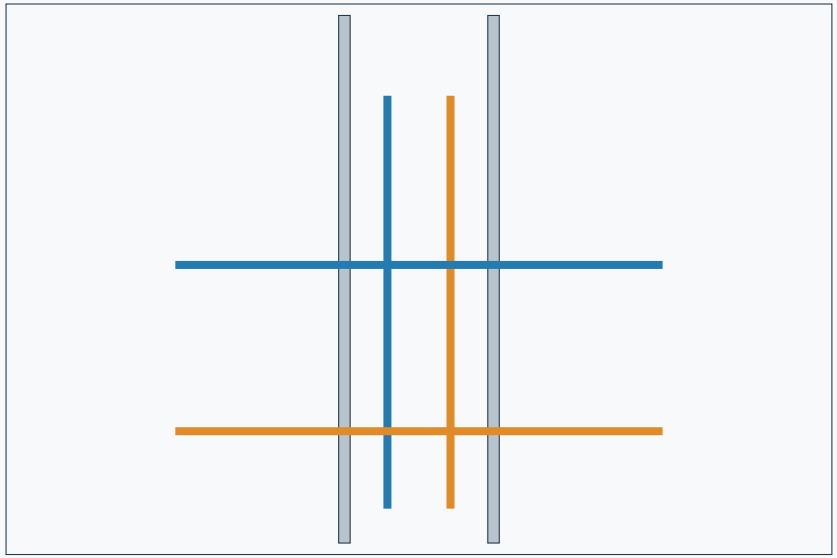
Sensitivity: total including allowance is 3.38 kW at -20 C, 4.18 kW at -30 C, 4.99 kW at -40 C and 5.79 kW at -50 C. Supply velocity is 1.33 m/s in main and branches; extract main is 2.00 m/s.

Assumptions: 4 people, 10 L/s/person, 0.15 ACH infiltration, fixed air properties; exposed roof and one end wall only. No internal gains or heat recovery. U-values are assumed. Thermal bridges, warm-up and humidity loads are excluded.

# D-03 | Route comparison and access

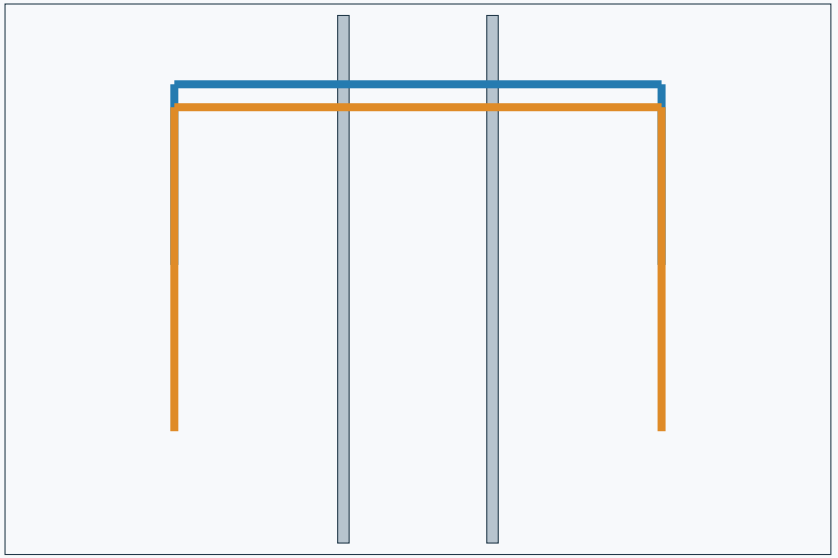
Two concept options | Equal terminal locations | Centerline estimates, not fabrication takeoffs

## A | Central corridor - selected



Supply: blue | Extract: orange

## B | Perimeter distribution



Supply: blue | Extract: orange

## Comparison basis

A: two 3.60 m mains + four branch spans totaling 8.496 m + 0.740 m drops = 16.436 m. B: two 4.248 m headers + four cabin legs totaling 8.800 m + 0.740 m drops = 18.036 m. Option B is about 1.60 m (9.7%) longer under this simplified routing basis.

Select A for shorter distribution and shared corridor maintenance access. B places mains over cabins and nearer the envelope. Length is the only quantified comparison; fitting losses, cost and acoustic performance are not inferred from length alone.

# R-01 | Study results and handover

Completed concept-study scope | Detailed marine engineering remains a separate phase

## Revision B resolves the draft geometry gaps

600 mm service zone; extract route raised 50 mm; 25 mm insulation volumes; enlarged partition openings; three support reservations; corridor heater reservation. The original native shell remains a separate early concept.

## Heating and equipment targets

At -30 C: 3.64 kW calculated, 4.18 kW including 15% allowance. Planning targets: 3.0 kW fresh-air coil, two 0.60 kW cabin heaters and one 0.30 kW corridor heater. These are capacity targets, not verified product selections.

## Verification performed

43 exported solids are valid. Duct and insulation envelopes clear modeled walls. Supply/extract insulation envelopes do not overlap. Minimum roof clearance is 164 mm. Zone heating reconciles to the revised total.

## Project boundaries

No claim of class approval, actual vessel integration or fabricated/tested HVAC performance. Deck/roof scantlings, fire dampers, intake/exhaust weather protection, controls, condensation, humidity and structural support sizing remain detailed-design work.

## Portfolio explanation

Describe this as an independent, AI-assisted HVAC concept study. Explain the boundary assumptions, heating split, route comparison and the insulation clash found during iteration. Do not present it as work performed for Seaspan or the Canadian Coast Guard.

## Reference context

Public Seaspan accommodation-design article; Genoa Polar Icebreaker project; ISO 7547 scope; IMO Polar Code overview. Exact links and all assumptions are retained in Design_Basis.txt. None supplies vessel-specific geometry or validates these assumed U-values.